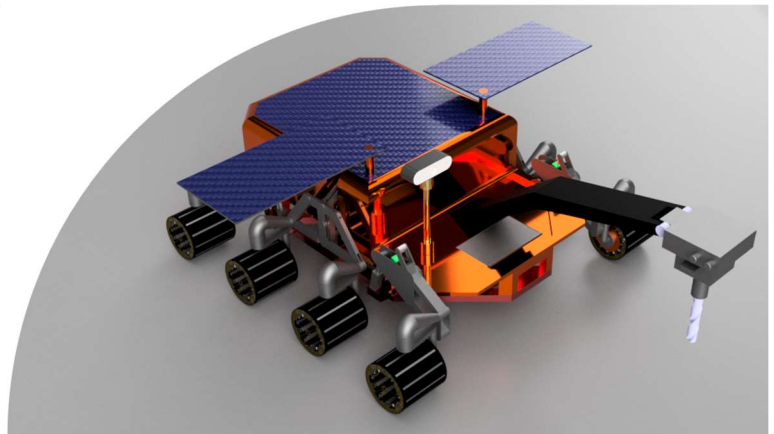


Adaptive Martian Rover Platform: Mechanisms Design and its Integration

I D F P R E S E N T A T I O N

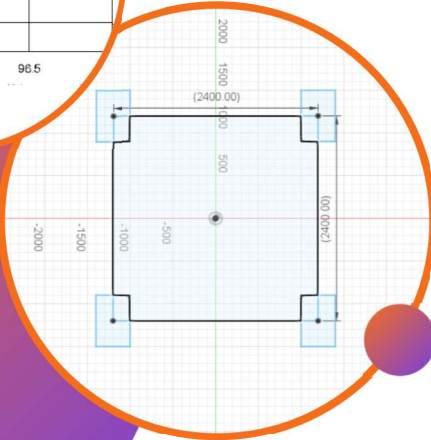
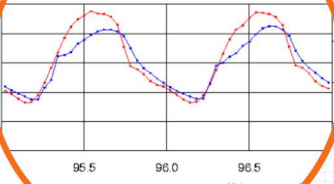
Problem Statement

- Core sampling up to a depth of 1 foot
- Performing of high intensity drilling operations on inclined surfaces with loose and rocky terrain
- Space-ready platform suitable as payload for GSLV-Mk.III
- Increase in speed and agility



Assumptions and Constraints

Mars Temperatures



- Track Width: 2.4m
- Wheelbase: 2.4m
- Vehicle Mass: 1500kg
 - Weight on Earth: 14715N (assuming $g = 9.81 \text{ m/s}^2$)
 - Weight on Mars: 5580N (assuming $g = 3.72 \text{ m/s}^2$)
- Avoid use of hydraulic and pneumatic actuation due to adverse environment conditions. Electric actuation is to be used.
- Low travel speed of the order 10 kph.
- Average temperature on mars: -70 C

Checklist

Compactible

Survives Launch and entry

Variable Ride Height

Face Martian Terrain & Weather

Compact for launch by occupying limited space.

Enduring launch and travel.
Safe landing of the rover on Martian terrain.

Achieving increased ground clearance along with low centre of gravity when required to tackle obstacles and steep inclines.

Can create preload for drilling.

Surviving Mars' extreme temperatures and rough terrain.

Key Areas of Rover Design



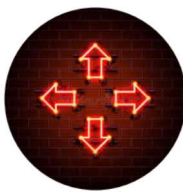
FRAME

Sustain high loads



SUSPENSION

Variable Ride Height



STEERING

Maneuverability



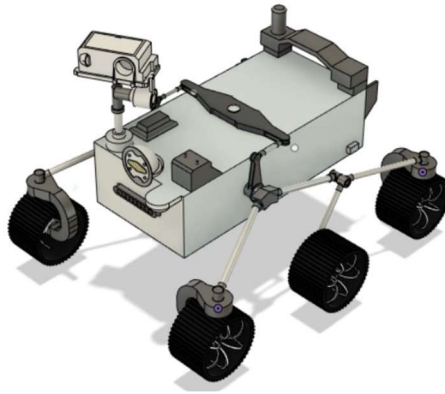
ARTICULATION

Maintain optimum contact

Rocker - Bogie

Pros

- Simple and most used.
- Load equally distributed on all wheels.
- Can overcome obstacles that are of the order of the size of a wheel.
- Stabilizes excessive pitching



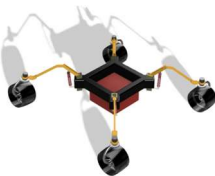
Cons

- Low operation speed.
- Pivot about a single line unable to support high vibrational loads.

Wheeled-leg Rover

Pros

- High mobility of legged systems on rough terrain.
- Square geometry - even distribution of loads.
- Variable ride height



Cons

- High actuator count.
- Cannot be controlled even if one actuator fails.
- Lowered traction
- Complex control of rover.

Six-Legged Rover

Pros

- High mobility.
- Suitable for rough terrain
- Symmetric geometry
- Presence of drilling unit in the centre allows for dynamic pitch control



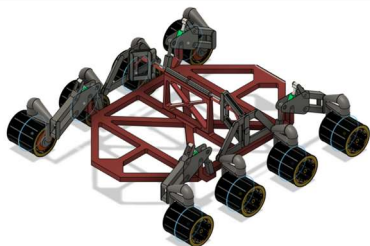
Cons

- High actuator count Direct and constant loads on actuators
- High power consumption
- Unsuitable for loose surfaces

Eight-Wheeled Rover

Pros

- Higher stability and traction than conventional systems
- Reduced toppling about a pivot
- Modularity of platform finds utility in other applications

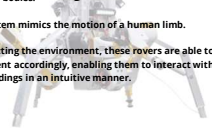


Cons

- Dependence on the holding torque and force of actuators
- Planar structure not resistant to out-of-plane impulses

Legged Systems

- These rovers can move on specific spots and can also tilt their bodies.
- This system mimics the motion of a human limb.
- By detecting the environment, these rovers are able to adjust their movement accordingly, enabling them to interact with their surroundings in an intuitive manner.



INFERENCES

- **The rover will be 8 wheeled due to the following reasons:**
 - Higher stability and traction
 - Reduced toppling about a pivot
 - Higher gradeability for a similar footprint
- **No dependence on actuators for height locking.**
- **Variable ride height will be achieved with lesser actuators.**
- **Straight line vertical movement of the rover wheels should be achieved**

Suspension System



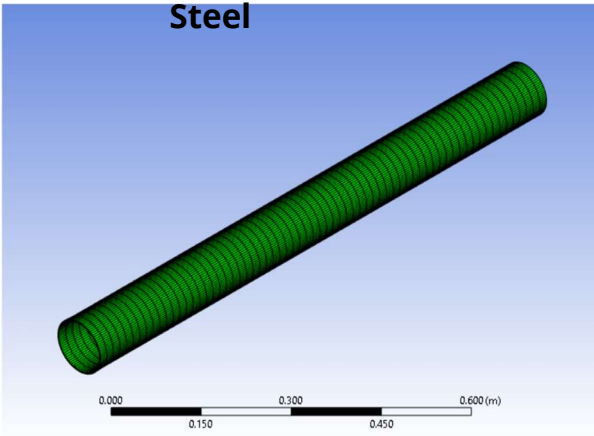
- Four suspension subassemblies
- The front and rear subassemblies will have common actuation
- $DOF = 1$
- Maximum travel: 355mm
- Pitch Range: -12.5° to $+12.5^\circ$
- Actuation by means of a DC motor via a 4.5mm pitch worm arrangement, closed loop control by means of linear potentiometer

Key Advantages

- No dependence on actuators for holding a particular configuration
- Minimized actuator count
- Use of 'low power' actuators
- Weight shift achieved by active pitch control to be implemented as preload for drilling operations
- Perfect straight line movement supports closed loop stability

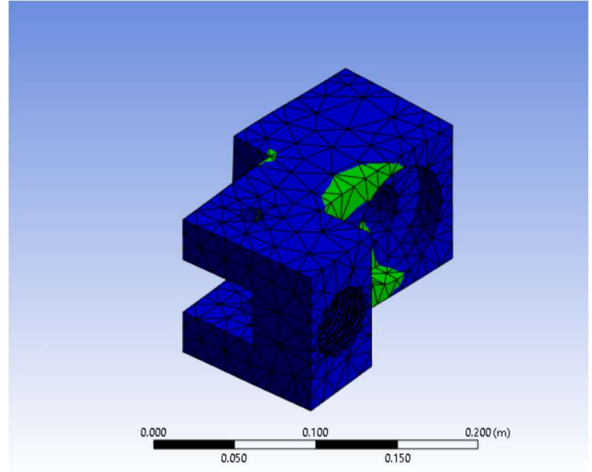
Material: Structural

Steel



Axle-Wishbone Mount:

- Minimum Safety Factor: 4.1
- Fatigue Life: 10^9 cycles



Axle Journal: 90mm dia., 5mm thick

- Minimum Safety Factor: 4.12
- Fatigue Life: 10^9 cycles

Linear Motion as a Solution

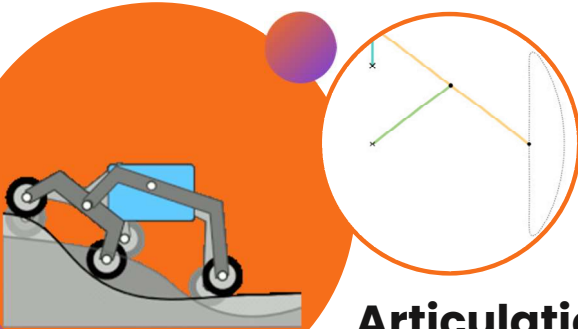
- Linear motion suspension is preferred as reduced moment from the obstacle is desired.
- Our proposed system is based on the **Chebyshev-Lambda** approximate straight line mechanism.
- Presence of an approximate straight line vertical motion ensures a vertical steering axis at all times.

Articulation

- The rover wheels must maintain contact with the ground, which needs a proper articulation assembly.

Need for an improved system:

- Current bogie systems are optimized for low speed and low to medium loads.
- Low coefficient of friction of obstacle or failure of mid-wheel actuator may cause bogie 'turn-over' at higher speeds.



Double Lambda Bogie

Advantages:

- Enforcing equal load distribution on both wheels.
- Improved articulation and increased operating speeds
- Maintains a uniform vertical steering axis for all wheel configurations.
- Maximum travel accommodated: 200mm.

Steering System

- The rover design has eight wheels, and thus requires steering for each wheel.

Need for an improved system:

- Increased number of actuators which increased power requirements and cost
- Different turning rates and angles for different wheels according to different situations, i.e. in place turning, crab walking etc.

Existing Solution

- All rovers in operation today have six wheels, and four corner wheels of these have a steering functionality.
- Since only four wheels need to be steered, vertically mounted steering motors are used to control the steer angle and steer rate of each of these four wheels independently.
- Implementing this on an eight-wheeled rover will increase the power requirements, as all eight wheels will require a steering functionality.



Proposed Solution

- Two successive wheels on each bogie arm have been coupled to a common steering mechanism.
- Actuator used: 300mm stroke length linear actuator
- Degree of Freedom: 1
- Maximum Steering Angle: 45° and
- Typical Dynamic Friction Torque on each wheel: 281.25Nm
- Maximum Linear Force Required from Actuator (~1.5 safety factor): 6kN
- Preferred Actuator: [Ewellix CAHB-31N](#)

Variable Configuration Steering

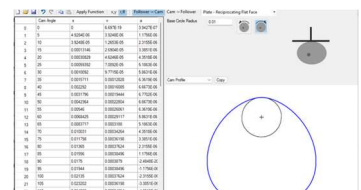
- The proposed steering system involves integration of two wheels in a common steering subassembly.
- This requires a system to switch the steering for in-place turning and turning along a curve scenarios, both requiring different steering angles and steering rates.
- A two-speed gearbox has been proposed to achieve this functionality, with a 1:1 and 7:17 gear ratio.

Two-Speed Gravity Assisted Gearbox

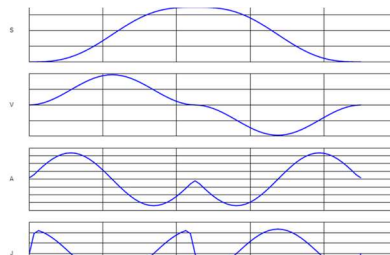
- A gearbox of two gear ratios is used, with a force (gravitational) closed cam and follower mechanism.
- The gear ratios are 1:1 and 7:17, giving the inner wheel a maximum steering angle of 45° and 26.56° respectively.
- The output gears have been provided with flanges. These ensure synchronized meshing at only the zero-position and prevent un-meshing when engaged.
- The cam is driven by an [MG996R](#) servo motor.

Cam Profile Design

- Application Used: Analytix Cams
- Parameters Set:
 - Initial Displacement: 0
 - Final Displacement: 35mm
 - Base Circle Radius: 10mm
- The following cam profile was obtained.



S-V-A-J Graphs obtained



Wheels and Tyres

- Current Mars rovers use monoblock aluminium alloy wheels.
- Due to low temperatures, rubber like materials cannot be used owing to glass-transition.
- Current Mars rovers face an issue for loose terrain traversal due to limited tyre technology.
- NASA's Spirit rover was rendered out of service because of such 'sand trap'.

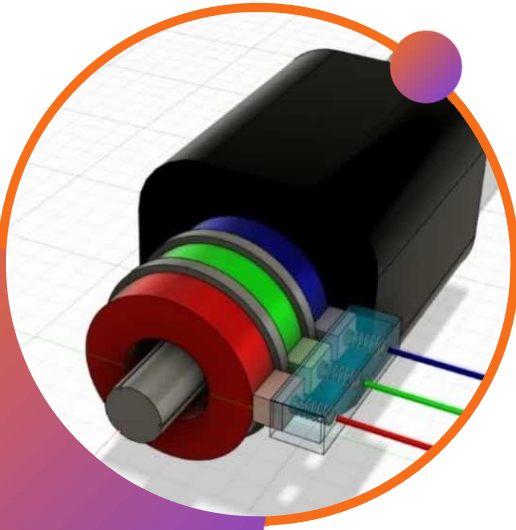


Proposed Retractable Paddle Wheel

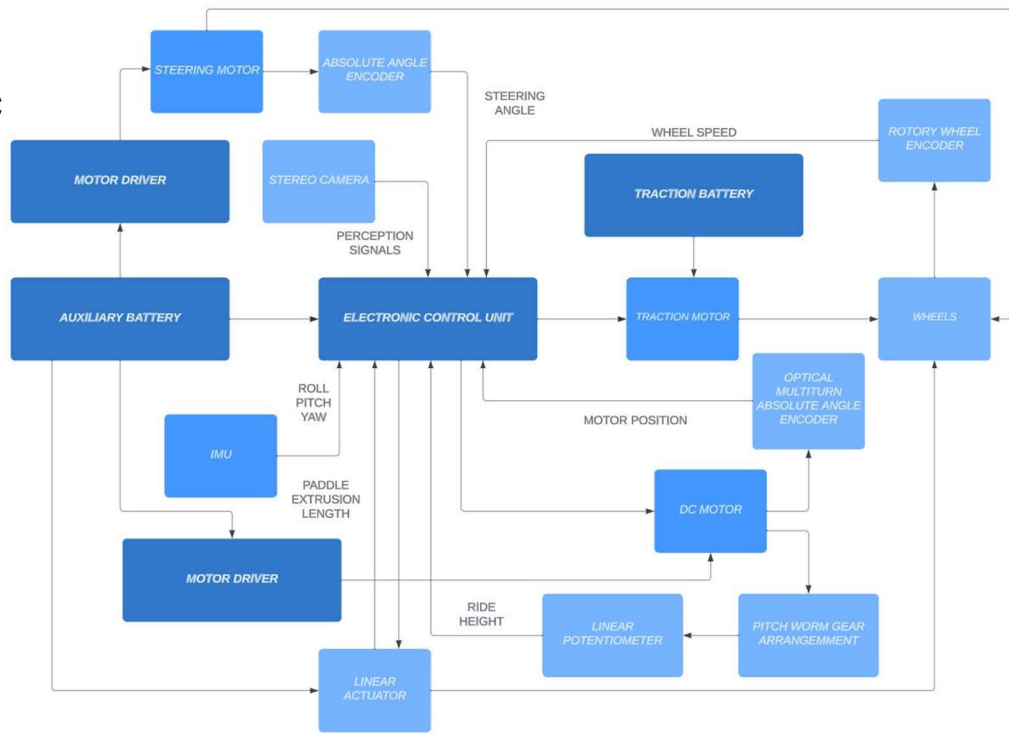
- Adding paddles to wheels will add traction in sandy terrain, but these fixed paddles will get damaged when traversing firm ground.
- Hence, a linear actuator based retractable paddle mechanism has been proposed.
- The wheel will consist of twelve such paddles, extendable to a maximum of 65mm.
- The spatial mechanism can be simplified to an 'elliptical trammel' mechanism with a single degree of freedom.

Actuator Integration

- A 150mm stroke length 6000N linear actuator shall be used.
- The actuator will be rotating about its axis with respect to the ground plane as it is rigidly attached to the rotating wheel subassembly.
- To ensure electrical connectivity of the actuator, its wires will be connected via a spring-loaded carbon brush. The wires will be passed through a slotted shaft to each of the rotating rings. These rings will act as a cam to the spring-loaded brushes (follower).



Control Schematic



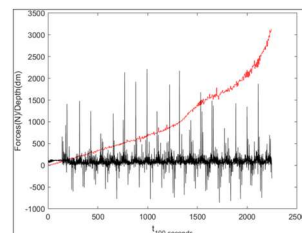
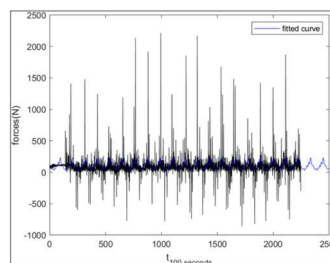
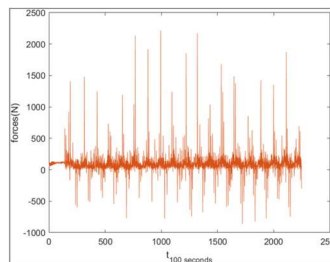
To understand the loads created due to high intensity core sample drilling, simulations were performed using a specialized drilling arrangement with assistance from Team Abhiyaan and MTL.

Data regarding the drilling forces and their variation with time and depth were recorded and plotted on graph.

Drilling Data Acquisition

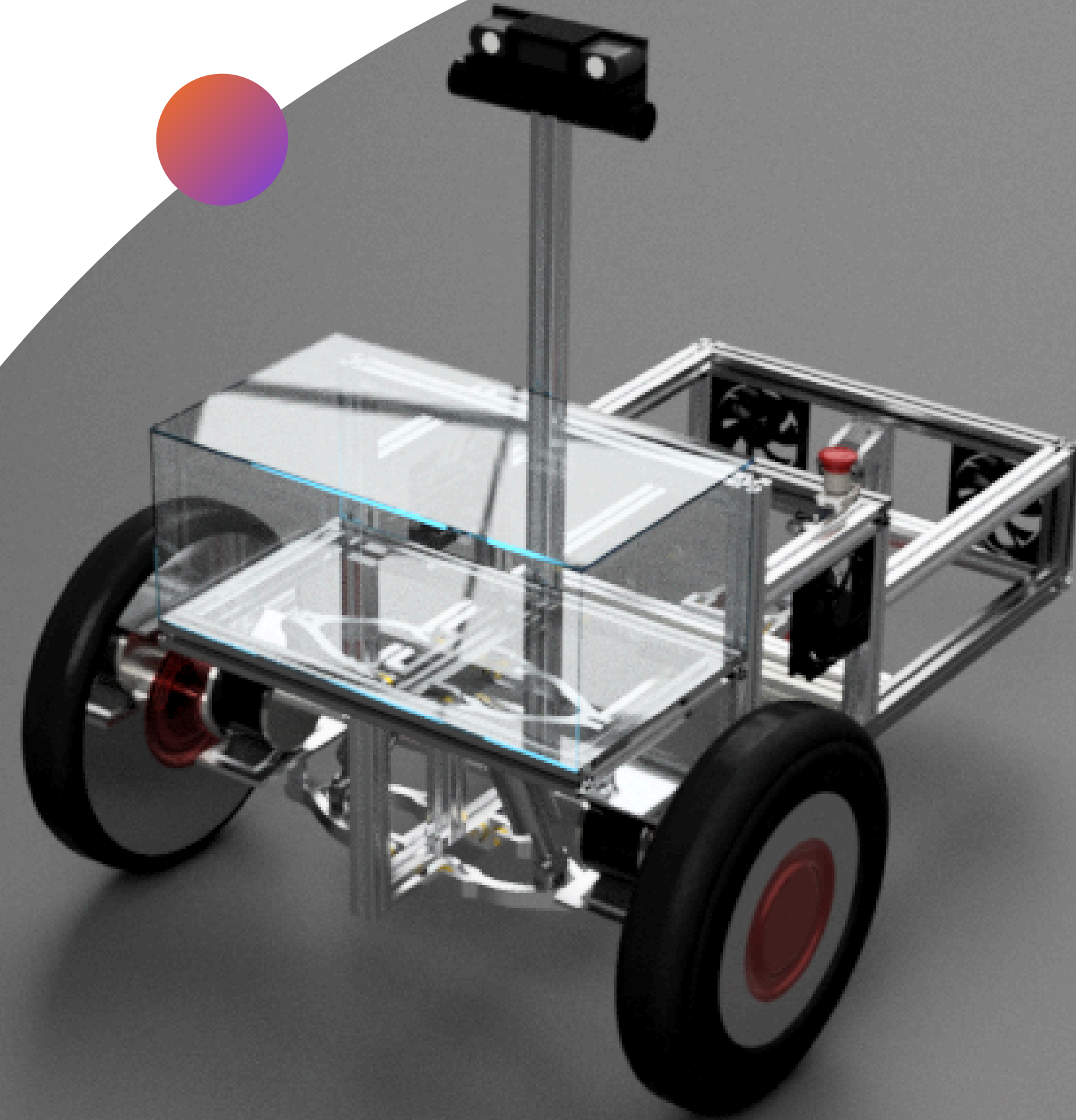
Inferences

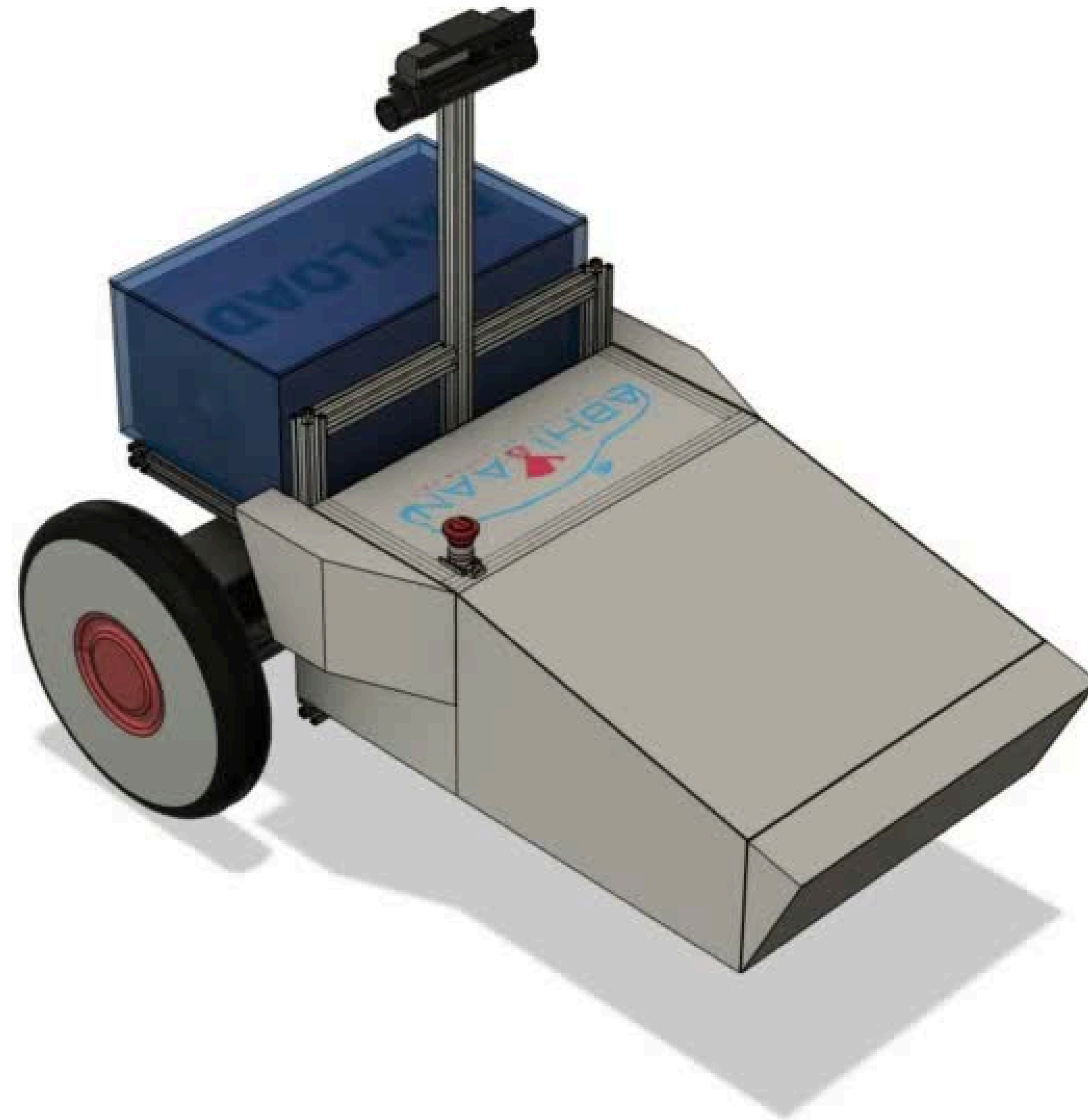
- The maximum positive peaks occur in the middle of the time duration due to the dominance of cracks in the rock.
- The drilling forces exhibit a uniform pattern with regular intervals of both positive and negative peaks due to the **percussive drilling** mechanism.
- Positive peaks represent the breaking down of hard rock by the bit, while negative peaks are generated when the drill bit pulls up rocks.



Autonomous Ground Vehicle

Mechanical Design



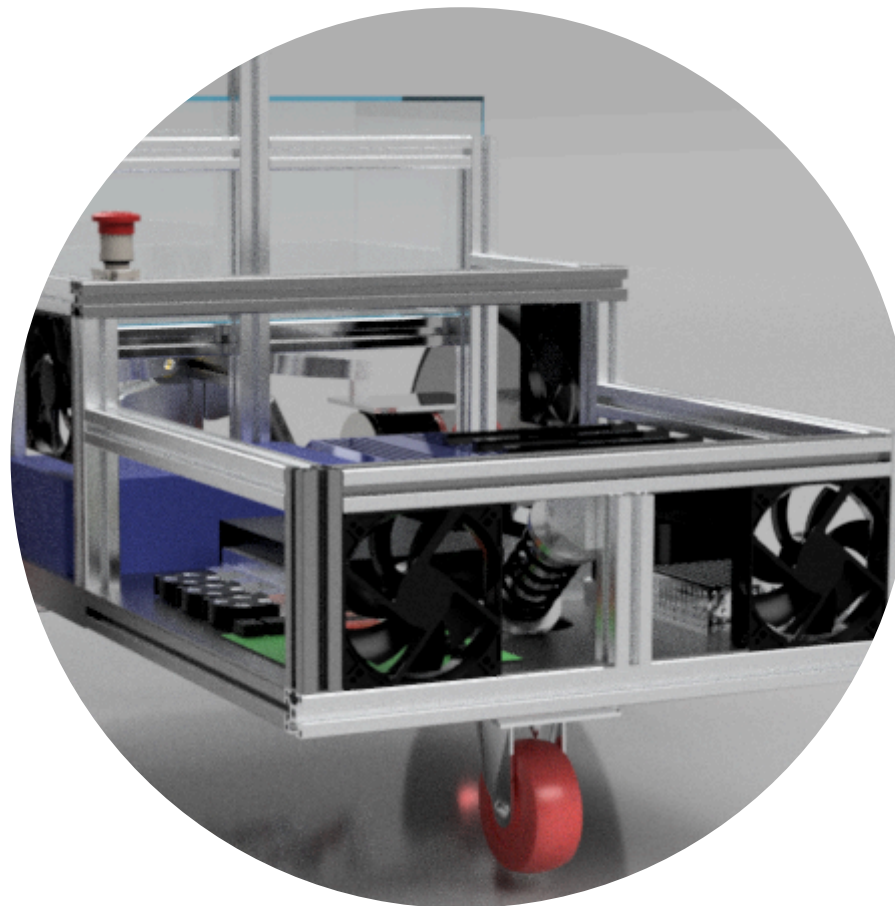


[For Exploded View Click Here](#)

Effective Innovations



**Independent
Suspension**



**Active Cooling
System**



**Improved
Accessibility**

Independent Suspension System

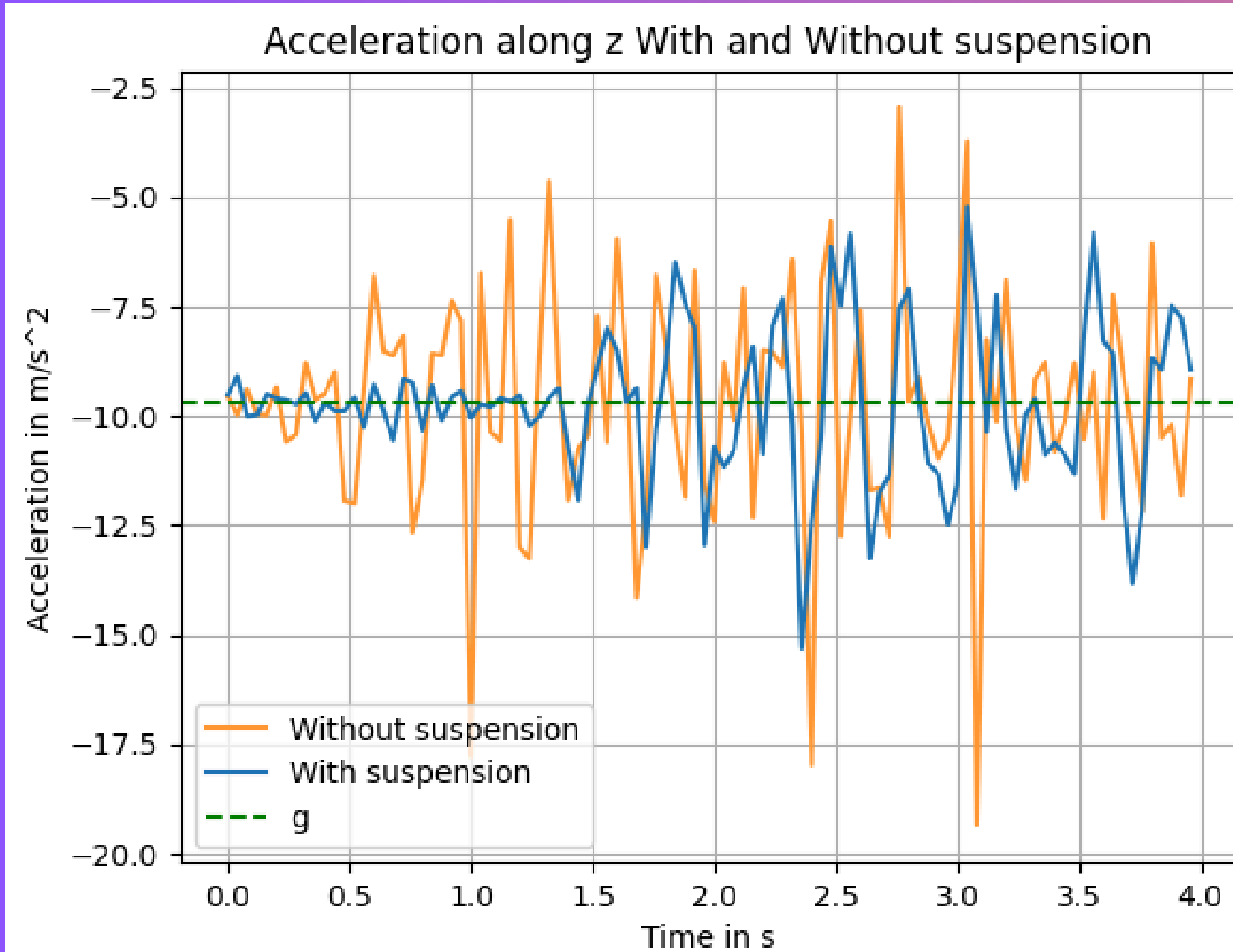
- **Drive Wheel Suspension**

- A double wishbone system setup is employed for the two front drive wheels
- Increased articulation and better ride quality

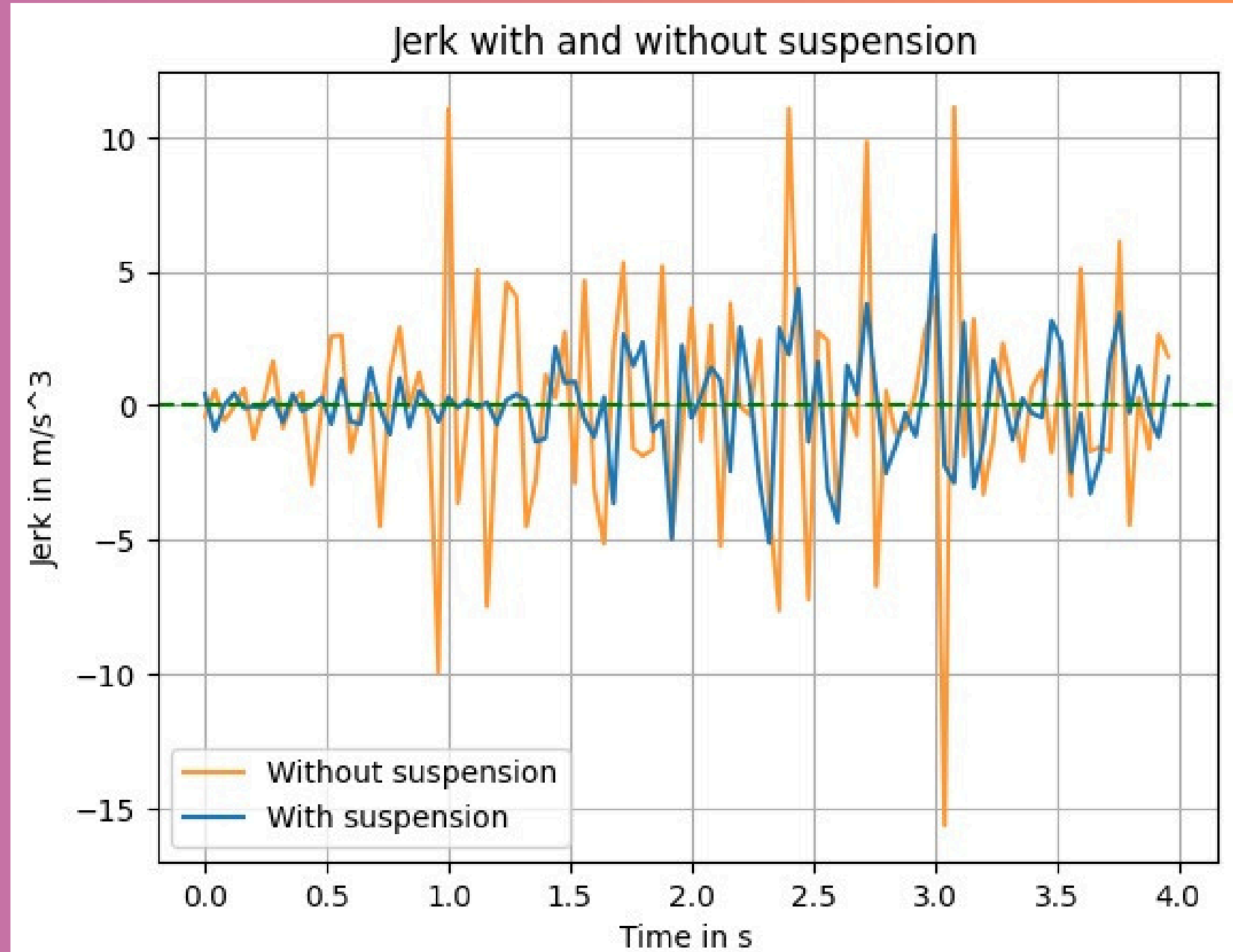
- **Castor Wheel Suspension**

- A trailing arm suspension system has been employed at the rear castor wheel
- Castor wheel is a point of high amplitude shocks due to a single contact-point
- Off the shelf spring-loaded castor wheels provide insufficient damping and wheel travel.

[For Suspension Animations Click Here](#)



z-Acceleration vs time



z-Jerk vs time

Active Cooling System

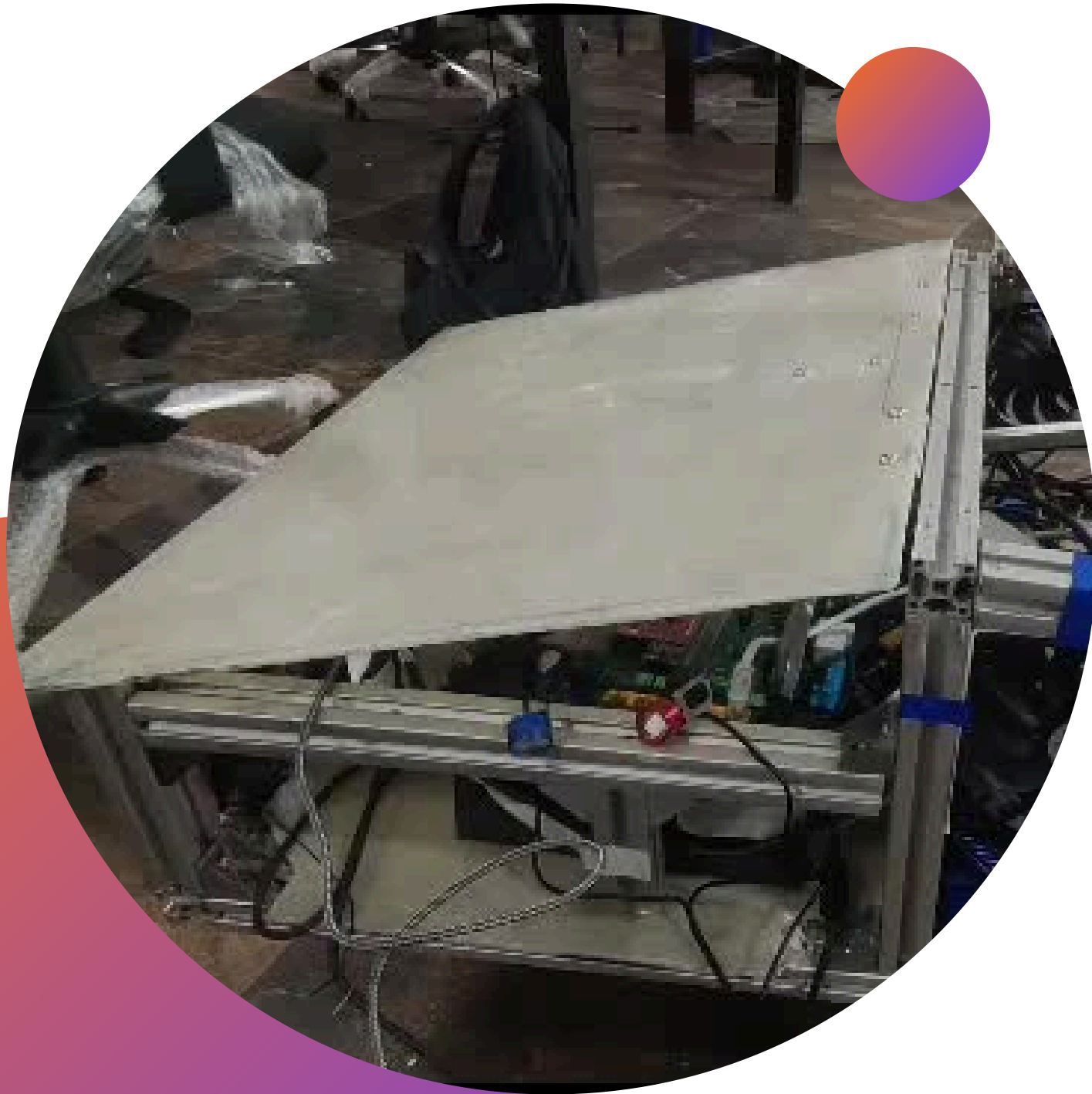
- **NUC External Cooling System**

- External cooling manifold for NUC
- Use of high pressure head radial fans
- Ability to lower the peak temperature by 4°C

- **Ventilation System for the Main Enclosure**

- Ventilation system installed to provide airflow to electronic components
- High volume flow rate axial flow rate fans have been installed on the surface
- About 200W of heat is dissipated during normal operation.

[For CFD Animation Click Here](#)



Improved Accessibility

Electronically Actuated Hatch - The hatch of the main compartment and the payload bay can be remotely operated to open and close at the push of a button, using a linear actuator and micro-servo motors

[For Hatch Actuation Video Click Here](#)